



Ontology-Guided Knowledge Graph Retrieval for Multi-Hop and Cross-Granularity Store Fulfillment Queries

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Abstract

Answering complex queries in store fulfillment, such as "What percentage of employee-assigned actions remain unresolved?" or "How many worklists for a specific product type were completed within a timeframe at each location?" requires precise, multi-hop reasoning across datasets with varying granularities. This paper introduces an ontology-based knowledge graph (KG) approach integrated with a structured text-to-Cypher generation pipeline, enabling accurate retrieval for such queries. Benchmarking against a robust hybrid search baseline combining BM25 and semantic search, our method demonstrates superior performance in addressing multi-hop and cross-granularity questions. Leveraging a KG schema designed to capture intricate relationships (e.g., (OrderLineItem)-[:INVOLVES_ACTION]->(Action)-[:INVOLVES]->(BatchProcess)-[:IS_COMPLETED_AT]->(Location)), we reveal universal patterns for constructing and querying highly relational data. This work highlights the transformative potential of ontology-driven KGs to improve reasoning, data aggregation, and decision-making, with broader implications for any domain requiring structured, multi-relational data analysis.

CCS Concepts

- Computing methodologies → Artificial intelligence → Knowledge representation and reasoning
- Information systems → Information retrieval → Information retrieval query processing

Keywords

Knowledge graph; multi-hop reasoning; ontology-driven retrieval; Large Language Model; text-to-Cypher; natural language processing; hybrid search; store fulfillment.

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1 Introduction and Motivation

Retail fulfillment involves interlinked data sources—purchase orders, shipping records, inventory, and worklist logs—that must be integrated for operational decision-making. Traditional retrieval methods struggle with multi-hop queries requiring structured reasoning, such as linking orders to fulfillment actions or tracking shipments across locations. While LLMs and hybrid search methods handle simple lookups, they fail when queries demand precise entity relationships or systematic aggregation [9, 11, 13].

Recent advancements in Text-to-SQL and GraphRAG provide structured retrieval via relational queries or embeddings and could be applied to fulfillment queries. However, Text-to-SQL operates within a fixed schema, lacking multi-relational modeling, while GraphRAG prioritizes embedding-based retrieval over explicit multi-hop execution. While viable alternatives exist, this study focuses on KG-based structural retrieval, evaluating its effectiveness in real-world fulfillment queries where explicit relationship traversal is essential.

This paper introduces a knowledge graph (KG) framework with a domain-specific ontology that models orders, actions, worklists, and locations. Applying an ontology-guided LLM to generate Cypher queries, we evaluate its ability to handle complex structural retrieval and identify key challenges such as Cartesian product errors, path misspecification, and filtering inconsistencies. We also evaluate KG retrieval in comparison to hybrid search (BM25 + semantic search), a powerful baseline that, while effective in some cases, struggles with systematic reasoning and cross-granularity aggregation. Our key contributions include:

- **Ontology-Driven KG for Fulfillment** – Applies KG beyond retail recommendation systems, enabling structured

retrieval for fulfillment operations through ontology-guided Cypher generation.

- **Structured KG Traversal for Relational Data** – Explores a graded approach to query complexity of structured relational data, providing a systematic evaluation of multi-hop reasoning in KG retrieval. Prior research has predominantly focused on KG construction from unstructured documents, with less emphasis on structured relational data.

In evaluating retrieval performance, we focus on queries joining up to three interrelated fulfillment data sources. A primary bottleneck is the occurrence of cross joins, where unintended pairwise combinations inflate intermediate results, leading to Cartesian product errors that reduce precision.

2 Related Work

Retail knowledge graphs have been widely used for recommendation systems, such as Amazon-KG's integration with DBpedia [10], and for e-commerce search, as demonstrated by Walmart's knowledge graph [1]. However, their application in fulfillment operations remains underexplored, particularly in contexts requiring ontology-driven, multi-hop reasoning.

Recent developments in retrieval-augmented generation (RAG) systems, including GraphRAG [2] and HybridRAG [7], leverage knowledge graphs by extracting nodes and relationships from unstructured documents rather than querying structured relational databases [4]. These systems predominantly rely on embedding-based retrieval, effective in various domains and supports multi-hop reasoning over unstructured data. However, they do not incorporate ontology-driven reasoning over

structured data, which is crucial for fulfillment tasks requiring explicit entity relationships and structured query execution.

Work on multi-hop retrieval over structured data has largely focused on optimizing table selection for SQL query generation, as seen in approaches such as SQL-PaLM [8]. While these methods enhance structured retrieval in relational database contexts, they remain constrained by fixed table schemas. Knowledge graphs offer an alternative approach by encoding entity relationships in an ontology, facilitating dynamic traversal and cross-granularity aggregation beyond the capabilities of table-based retrieval [5, 6].

Xu et al. (2024) from LinkedIn [12] explored the use of knowledge graphs in RAG for customer service and ticketing, employing Cypher for structured retrieval. Their approach demonstrates the viability of structured query techniques within RAG but does not explicitly address cross-table retrieval, granularity, or aggregation—elements that are particularly relevant for complex multi-hop queries over structured data.

3 Methodology

3.1 Knowledge Graph Construction

We standardize shipment, order, worklist, and inventory data from multiple sources, ensuring consistent product identifiers. The resulting Neo4j knowledge graph comprises 100,000 nodes and 200,000 relationships, structured with a concise ontology defining key entities such as GuestOrder, OrderLine, Product, Inventory, and SubLocation, along with relationships like CONTAINS, HAS_INVENTORY, and IS_PART_OF. The dataset represents a filtered snapshot of fulfilled orders in a selected store, covering 20,000 orders.

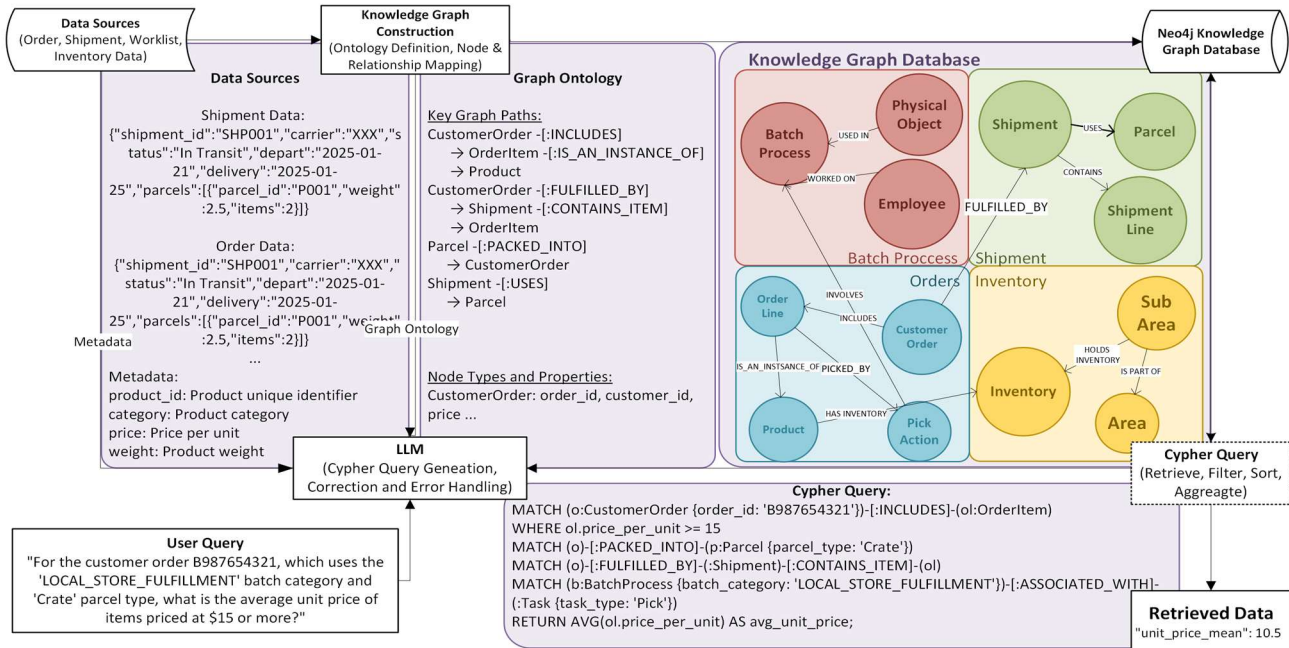


Figure 1. System Design for Knowledge Graph-Based Store Fulfillment Query Processing (The figure is a simplified representation of the knowledge graph, omitting select node types and edges for clarity. All data shown is anonymized or synthetically generated.)

3.2 LLM Cypher Generation and Query Architecture

GPT-4o translates natural language questions into Cypher using a three-step process: initial generation (producing ontology-aligned queries), ontology correction (fixing label and relationship mismatches), and attribute validation (ensuring correct property references). A fuzzy-matching loop resolves ambiguities, such as location names, before execution, returning either detailed or aggregated results, such as container counts.

3.3 Synthetic Data Generation

To systematically evaluate multi-hop query performance, we merged raw tables (orders, shipments, worklists) into a unified dataset. An LLM-based pipeline structured queries by selecting aggregation, sorting, and filtering criteria based on predefined topics and columns, enforcing n-hop complexity, where each hop corresponds to a relational join across different data granularities. It then produced natural language questions over those structured queries to simulate real-world retrieval scenarios. The queries were human-verified to ensure they accurately reflected the intended difficulty level.

By constraining the columns used, this pipeline forced multi-hop dependencies, emphasizing the KG's ability to traverse structured relationships rather than relying on general linguistic understanding. For example, a 1-hop query might ask: "For shipments processed with the X batch type, using Box containers in the Y area, what is the total number of shipment lines?" Here, the KG must join worklist (batch type) data with shipment details to compute the requested metric.

3.4 Query Complexity and Validation

We evaluated 542 test cases, varying in complexity across two dimensions: the number of tables joined, and the type of aggregation performed. The queries ranged from 0-hop, which involved retrieving data from a single table to 1-hop, where two tables were joined, and 2-hop, where three tables were joined. Query complexity increased with aggregation type. Self-key aggregation operated within a single table using primary keys. Joined-key aggregation used foreign key joins while aggregating on the base table and/or the joined table's primary keys. Cross-granularity aggregation required hierarchical alignment of non-primary keys across tables. The most complex cases involved multiple joins and cross-table aggregation on non-primary keys, requiring structured traversal. An LLM-based evaluator validated correctness by checking whether the retrieved outputs matched ground-truth data after column mapping.

3.5 Hybrid Search Baseline

For comparison, we created a baseline using Elasticsearch, where each row from the orders, shipments, and worklists tables was stored as a JSON chunk. At query time, GPT-4o was provided with foreign and primary key relationships to manually link retrieved chunks using a hybrid approach that

combined BM25 for lexical retrieval and semantic embeddings with text-embedding-3-large. The weighting between lexical and semantic retrieval was optimized through hyperparameter tuning. While this baseline performed well on simpler queries, it struggled with multi-hop reasoning and cross-granularity aggregation due to its lack of explicit relational structure.

4 Experiment Results

4.1 Performance Advantage of Ontology-Based KG Over Hybrid Search

Table 1 demonstrates that as query complexity increases, the ontology-based KG consistently and significantly outperforms hybrid search by a substantial margin. In 0-hop queries, which involve retrieving data from a single table without joins, the KG achieves a slight advantage (+2.5%). However, as queries require more complex relational traversal—1-hop involving a single join and 2-hop requiring two joins—the performance gap widens considerably, with KG surpassing hybrid search by 37.95% and 21.31%, respectively. The most pronounced decline in hybrid search accuracy occurs at 1-hop, where it drops to 3.45% from 76.5%, indicating its inability to effectively retrieve and align relational data across tables. In contrast, the KG maintains significantly higher accuracy across all levels of complexity, reinforcing its superiority in structured multi-hop retrieval.

The ablation study in Table 1 assesses the impact of error handling and ontology guidance on KG retrieval performance. Error handling includes attribute correction, Cypher path refinement, and query-time validation to mitigate execution failures. Its removal leads to a progressive decline in accuracy, with reductions of 8.0% at 0-hop, 12.1% at 1-hop, and 13.9% at 2-hop, demonstrating its critical role in preventing filtering inconsistencies and path misalignment. The removal of ontology guidance results in complete retrieval failure (0% accuracy across all query complexities), underscoring its necessity for structuring relationships and ensuring valid Cypher generation. These findings highlight the essential role of ontology-driven reasoning in multi-hop retrieval and emphasize the importance of robust error handling in maintaining query precision.

Table 1. Query Retrieval Accuracy with Ablation Study. All values are percentages. Hybrid = hybrid search; Full KG = full knowledge graph; KG Gain = improvement over Hybrid; -Err = KG without error handling; -Onto = KG without ontology guidance; Δ = accuracy drop from full KG.

# of Hop	Hybrid	Full KG	KG Gain	- Err	Δ -Err	- Onto	Δ -Onto
0	76.5	79.0	2.5	71.0	-8.0	0.0	-79.0
1	3.45	41.4	38.0	29.3	-12.1	0.0	-41.4
2	2.29	23.6	21.3	9.71	-13.9	0.0	-23.6

The query complexity in Table 2 is graded based on the number of tables joined and the type of aggregation performed, as detailed in Section 3.4. Table 2 highlights the KG's ability to

handle structured, multi-hop retrieval even as query complexity increases. Unlike traditional search methods, the KG maintains higher accuracy across all aggregation levels, excelling in hierarchical relationships and structured joins. Its advantage is most evident in cross-table queries, where hybrid search accuracy drops to near-zero ($\leq 3.64\%$), while the KG retains substantially higher accuracy despite some performance decline.

The KG consistently outperforms hybrid search, though the margin varies by query type. The largest gain (+42.9%) occurs when retrieving data from a related table, while cross-granularity aggregation via a single join shows a +33.8% improvement, highlighting the KG's strength in hierarchical alignment. Simpler tasks, like single-table aggregation, see a modest +2.5% boost. For two-join queries, the KG maintains a clear lead (+29.1% and +17.0%, depending on key type), further showing its advantage in structured retrieval. Hybrid search accuracy remains low across complex queries due to reliance on keyword matching and embeddings over structured joins.

Overall, the KG shows a clear advantage in structured, multi-hop retrieval, particularly for queries spanning multiple levels of detail. While its performance gain drops in more complex queries due to scaling challenges like Cartesian products and ambiguous path specifications, it remains superior to hybrid search. Hybrid search performs well for simple aggregations but struggles with explicit joins and hierarchical reasoning, reinforcing the KG's effectiveness in store fulfillment queries.

Table 2. KG Vs. Hybrid Query Performance by Complexity

Aggregation Type	KG Acc (%)	Hybrid Acc (%)	KG Gain (%)	Explanation
No Join	79.0	76.5	2.50	Aggregation within a single table, no joins.
1-Join (Primary Key)	44.0	1.10	42.9	Join one table; aggregate on base-table primary keys.
1-Join (Cross-Granularity)	36.5	2.70	33.8	Join one table; aggregate on non-primary keys.
2-Joins (Primary Key)	32.7	3.64	29.1	Join two tables; aggregate on joined table primary keys.
2-Joins (Cross-Granularity)	18.0	1.00	17.0	Join two tables; aggregate on non-primary keys across levels.

The query illustrated in Figure 1 demonstrates the knowledge graph's advantage over hybrid search in handling structured retrieval across multiple fulfillment levels. The query for customer order B987654321 requires filtering items at the order item level based on price, associating them with the correct batch category, and computing the average price while ensuring alignment with parcel constraints. This introduces a multi-granularity challenge, as order items are linked to customer orders, packed into parcels, and processed under batch

operations that govern fulfillment. The KG approach effectively handles this complexity by explicitly linking CustomerOrder $\rightarrow$ OrderItem $\rightarrow$ Parcel $\rightarrow$ Shipment $\rightarrow$ BatchProcess $\rightarrow$ Task, allowing for precise filtering and aggregation. The Cypher query in Figure 1 explicitly enforces these constraints.

Hybrid search retrieves customer order B987654321 and some shipment lines containing the specified parcel type but fails to retrieve all necessary shipment records, leading to incomplete information. Additionally, it does not capture the required batch category, leaving a critical condition unverified. Without this data, the retrieved results remain insufficient for the LLM parser to generate an accurate aggregation. By enforcing explicit constraints at each step, the knowledge graph ensures that all conditions are correctly applied, resulting in more precise and reliable aggregation compared to hybrid search, which lacks explicit validation mechanisms.

4.2 KG Retrieval Bottlenecks

Despite its advantages, KG retrieval faced several failure modes, especially as query complexity increased. The most common issue was Cartesian product explosion (20.3%), where missing relationship patterns led to excessive cross-joins and inflated counts. For instance, independently retrieving order items and batch processes without enforcing structural constraints caused unintended pairings, overcounting fulfillment quantities. Despite prompt instructions, LLM-generated Cypher queries inconsistently applied constraints, especially in multi-branch queries, worsening aggregation errors.

Path misspecification (23.4%) frequently resulted in empty results due to incorrect traversal, while inconsistent filtering (3.13%) led to overcounting. LLM hallucinations (7.81%) introduced nonexistent nodes or relationships, preventing execution, and natural language understanding errors (6.25%) caused attribute misalignment, leading to incorrect entity selection. These challenges highlight the need for improved ontology representations and more precise LLM-driven Cypher generation. Addressing inconsistencies in constraint enforcement and enhancing query structure alignment will be critical to improving retrieval accuracy in multi-hop queries.

4.3 Conclusion and Future Work

This study shows that ontology-based KGs outperform hybrid search in structured, multi-hop retrieval for store fulfillment data system. By leveraging explicit entity relationships, KG retrieval enables more accurate multi-table structured retrieval, whereas hybrid search struggles with increasing query complexity due to reliance on lexical matching and embeddings. Future work will focus on refining query post-processing techniques to prevent inflated result sets caused by Cartesian Products. Fine-tuning Cypher generation will improve schema alignment and reduce retrieval errors. Additionally, scaling to more complex queries will further assess performance in higher-order reasoning tasks, enhancing the robustness of ontology-driven retrieval systems for enterprise applications.

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